

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]

Tap amplitudes (pixels): [146.40013661195812, 127.00393694685216, 117.00427342623003, 127.7810627597063, 131.36590120727678, 141.65097952361643, 117.92370414806346, 122.44182292011173, 120.0, 126.21014222319853, 118.92854997854805]



Based on the extracted features provided, here is the analysis and scoring for the MDS-UPDRS Item 3.4 Finger Tapping test.

Analysis of Extracted Features

1. Rhythm and Speed (Periods)

- **Data:** [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]
- **Observation:** The tapping speed is inconsistent. There is significant variance between taps (e.g., accelerating to 155 frames, then slowing to 203 frames, then back to 171).
- **Trend:** There is **no consistent slowing** (fatigue) trend. The final tap (170) is actually faster than the first tap (189). However, the fluctuations (such as the jump from 155 to 203) suggest minor hesitations or breaks in the regular rhythm.

2. Amplitude (Pixels)

- **Data:** [146.4, 127.0, 117.0, 127.8, 131.4, 141.7, 117.9, 122.4, 120.0, 126.2, 118.9]
- **Observation:** The subject starts with the largest amplitude (146.4). There is an immediate drop in

amplitude by the 3rd tap (117.0).

- **Trend:** Crucially, the amplitude **recovers** mid-sequence (climbing back up to 141.7 at tap 6). This recovery rules out the pathological "Sequence Effect" (progressive decrementing) typically seen in scores 2, 3, or 4.
- **Conclusion:** While the subject fails to maintain the initial maximal amplitude throughout the entire sequence (ending around ~120 vs starting at ~146), the ability to recover amplitude mid-test suggests the impairment is minimal.

Scoring Rationale

- **Not Score 0 (Normal):** The rhythm is not perfectly stable, and the amplitude is not consistently maintained at the maximal level established by the first tap.
- **Not Score 2, 3, or 4:** The subject does not exhibit "Mild," "Moderate," or "Severe" slowing, nor do they show a progressive, unrecoverable decrement in amplitude. The amplitude bounces back up in the middle of the test.
- **Matches Score 1 (Slight):** The data fits the criteria for "Slight" impairment due to:
 - **1a:** The regular rhythm is broken with interruptions/fluctuations (period variance).
 - **General Performance:** The inability to strictly maintain the amplitude of the first tap indicates a "slight" deviation from normal, even though functional recovery occurred.

Score: 1